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EXAMINER
DIXON, ANNETTE FREDRICKA

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3785	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/232,058	Applicant(s) BARLOW et al.	
	Examiner ANNETTE F DIXON	Art Unit 3785	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 9/19/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 11-30 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 11-30 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 8/9/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☑ Certified copies of the priority documents have been received in Application No. 17426362.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date ALL IDS.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. Primary Examiner acknowledges Claims 11-30 are pending in this application, with Claims 11-30 having been newly added, and Claims 1-10 having been cancelled by preliminary amendment on September 19, 2023.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 103

3. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

4. Claims 11-28 and 30 are rejected under 35 U.S.C. 103 as being unpatentable over Ovizinsky et al. (2014/0102456) in view of Hamblin (2,452,047).

As to Claim 11, Ovizinsky discloses a positioning and stabilizing structure (Figures 6-15) to provide a force to hold a seal-forming structure (510 as shown in Figure 6, "As illustrated, a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end

520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit." Para 0117) in a therapeutically effective position on the patient's head for positive pressure airway treatment of sleep disordered breathing ("The present technology relates to air delivery conduits used in Positive Airway Pressure (PAP) systems for treatment, e.g., of Sleep Disordered Breathing (SDB) with Continuous Positive Airway Pressure (CPAP) or Non-Invasive Positive Pressure Ventilation (NIPPV)." Para 0002), the positioning and stabilizing structure (Figures 6-15) comprising: at least one gas delivery tube (x20, shown as 520 in Figures 6 and 7 and 620 in Figures 8-15, "FIGS. 8 to 15 illustrate an air delivery conduit 620 according to another example of the disclosed technology. In this example, the air delivery conduit 620 includes first and second conduit portions 621, 622 that cooperate to form the conduit, i.e., first conduit portion provides a portion of the conduit circumference and the second conduit portion provides the remaining portion of the conduit circumference." Para 0186) to deliver a flow of air to an entrance of the patient's airways via a seal forming structure (510 as shown in Figure 6, "As illustrated, a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit." Para 0117), each gas delivery tube (x20) being constructed and arranged to contact, in use, at least a region of the patient's head superior to an otobasion superior of the patient's head, each gas delivery tube (x20) comprising a tube wall (defined as the combination of 621 and 622, "In this example, the air delivery conduit 620 includes first and second conduit portions 621, 622 that cooperate to form the

conduit, i.e., first conduit portion provides a portion of the conduit circumference and the second conduit portion provides the remaining portion of the conduit circumference." Para 0186) defining a hollow interior (defined by the central region of 620) through which air is able to flow to the seal forming structure (510 as shown in Figure 6), each gas delivery tube (x20) comprising: a patient contacting portion (one of 620/621, "first and second conduit portions 621, 622 that cooperate to form the conduit" Para 0186) configured to contact the patient's body, the patient contacting portion (one of 620/621) comprising a textile material and/or foam material (655, "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric." Para 0187), and an air impermeable material (650, "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric." Para 0187; **also see**: "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622. During the thermoforming process, the laminate adheres to the fabric giving it air impermeable properties. The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit. In use, such a conduit keeps its form in the absence of external force. However,

the conduit is also collapsible and, when sufficient external force is applied, the conduit collapses. The form holding and collapsible features are particularly useful in the configuration shown in FIG. 6 which includes two conduits 520, one on each side of the patient's head. While both conduits maintain their form in the absence of external force, when the patient is lying in bed and turns to one side, the respective conduit conveniently collapses, thus improving patient comfort. The remaining conduit maintains the air supply to the patient interface 510. The conduit may also be air impermeable." Para 0188 and "In a further alternative, the textile or fabric may be sealed by means other than a film laminate. For example, the textile or fabric may be sealed by spraying on a polymeric substance, such as silicone, or a powder that is then heated to congeal and create an impermeable barrier." Para 0190) provided to the textile material and/or foam material (655) to make the patient contacting portion substantially air-impermeable; a non-patient contacting portion (other of 620/621, "first and second conduit portions 621, 622 that cooperate to form the conduit" Para 0186) configured to contact the patient's body, the non-patient contacting portion (other of 620/621) comprising a textile material and/or foam material (655, "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric." Para 0187), and an air impermeable material (650, "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric." Para 0187; **also**

see: "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622. During the thermoforming process, the laminate adheres to the fabric giving it air impermeable properties. The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit. In use, such a conduit keeps its form in the absence of external force. However, the conduit is also collapsible and, when sufficient external force is applied, the conduit collapses. The form holding and collapsible features are particularly useful in the configuration shown in FIG. 6 which includes two conduits 520, one on each side of the patient's head. While both conduits maintain their form in the absence of external force, when the patient is lying in bed and turns to one side, the respective conduit conveniently collapses, thus improving patient comfort. The remaining conduit maintains the air supply to the patient interface 510. The conduit may also be air impermeable." Para 0188 and "In a further alternative, the textile or fabric may be sealed by means other than a film laminate. For example, the textile or fabric may be sealed by spraying on a polymeric substance, such as silicone, or a powder that is then heated to congeal and create an impermeable barrier." Para 0190) provided to the textile material and/or foam material (655) to make the non-patient contacting portion substantially air-impermeable; and at least one seam (defined by the stacked configuration as shown in Figures 13 and 14, "The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit." Para 0188) to join

the patient contacting portion (one of 620/621) and the non-patient contacting portion (other of 620/621).

Yet, Ovizinsky **does not expressly disclose** the configuration "wherein the at least one seam comprises rigidising elements embedded within or layered on the seam to provide rigidity to the gas delivery tube."

Hamblin teaches the construction of an air delivery conduit constructed of a textile ("The present invention covers improvements in the means of construction of cloth and treated fabric hose used mostly as portable ventilating hose." Column 1, Lines 1-10; **also see**: "Cloth hose may be made up by methods described herein, either with or without wire stiffening as will be further explained herein." Column 1, Lines 5-20) suitable for the conveyance of fluids (e.g. air) having a first layer (one of the left or right planar butting structure as seen in Figures 2-4, 6, and 7) bonded to a second layer (one of the left or right planar butting structure as seen in Figures 2-4, 6, and 7) to form a seam ("seam" as seen in Figures 2-4, 6, and 7, explicitly with respect to Figures 1 and 2: "The method of construction of the present invention on ventilating hose consists in the use of one or more strips of cloth of relatively narrow width with the edges sewed together so that the seam runs longitudinally and circumferentially forming a helical seam as shown in Fig. 1 of the drawings clearly. The pitch of the seam will depend on the width of the strip selected for the cloth and the diameter of the hose. The seams may be sewed with any suitable type of stitching, or any other suitable means like riveting, etc., may be used." Column 1, Lines 50-60; with respect to Figure 3, "An alternate type of lap seam construction is shown in Fig. 3, in which the wire stiffening 10 is secured by turning edge 16 of strip I 1 back over wire 15. With some types of cloth or rubberized fabric, the "single strip" type of

construction shown in Figs. 1, 2 and 3 may cause a slight difference in diameter at the opposite ends, if the hose is long, unless special precautions are taken in manufacture. This may generally be accomplished by a tension device (not shown) on the sewing machine used in the fabrication of the hose." Column 2, Lines 5-25; with respect to Figure 4: "The tendency for a difference in diameter at opposite ends of the hose may be avoided by the use of a seam with edges of strip 18 turned up and stitched together at 19 as shown in Fig. 4." Column 2, Lines 20-30; with respect to Figure 6: "Another type of construction (Fig. 6) which may be used to avoid a difference in diameter of the hose is to provide two strips of material 22, 23 with both the "forward" and "after" edges of one strip 23 always lapping over the edges of the other strip 22, as shown in Fig. 6, where the edges 28a and 281b of strip 23 lap over the edges of strip 22. The wire stiffening for this type of construction (Fig. 6) will consist of two continuous wires 24 and 25." Column 2, Lines 25-40; and with respect to Figure 7: "Fig. 7 covers another form of the seam somewhat similar to that shown in Fig. 4, in which the hose material 28 is divided with two upstanding ends 21, 21 which may be used stitched at 28, 29 over which is mounted a cover strip 30. The strip 80 may be secured by an additional stitching at 28 if desired or secured in any other suitable manner. In this type the wire 30a may be used or omitted as found more desirable." Column 2, Lines 35-50).

Regarding the remaining limitations, Hamblin teaches the implementation of rigidizing element ("wire" as seen in Figures 2-4, 6, and 7, explicitly with respect to Figures 1 and 2: "In Fig. 1 is shown the hose made up with a single strip of cloth 10 wound helically, with the after edge 11 of each turn of the cloth lapped on and sewed to the forward edge 12 of the preceding "turn" of the cloth strip. Where wire stiffening is used with this type of construction to prevent

transverse collapse of the hose, the wire 13 is sewed between the lap of the turns of cloth, as shown in Fig. 2, with double stitching 14, 14." Column 2, Lines 1-10, with respect to Figure 3, "An alternate type of lap seam construction is shown in Fig. 3, in which the wire stiffening 10 is secured by turning edge 16 of strip I 1 back over wire 15. With some types of cloth or rubberized fabric, the "single strip" type of construction shown in Figs. 1, 2 and 3 may cause a slight difference in diameter at the opposite ends, if the hose is long, unless special precautions are taken in manufacture. This may generally be accomplished by a tension device (not shown) on the sewing machine used in the fabrication of the hose." Column 2, Lines 5-25, with respect to Figure 4; "The tendency for a difference in diameter at opposite ends of the hose may be avoided by the use of a seam with edges of strip 18 turned up and stitched together at 19 as shown in Fig. 4. Where wire stiffening is used in this type of construction the wire 20 is secured by binding strip 21. This binding strip may be used whether or not the wire stiffening is provided." Column 2, Lines 20-30; with respect to Figure 6: "Another type of construction (Fig. 6) which may be used to avoid a difference in diameter of the hose is to provide two strips of material 22, 23 with both the "forward" and "after" edges of one strip 23 always lapping over the edges of the other strip 22, as shown in Fig. 6, where the edges 28a and 281b of strip 23 lap over the edges of strip 22. The wire stiffening for this type of construction (Fig. 6) will consist of two continuous wires 24 and 25." Column 2, Lines 25-40; and with respect to Figure 7: "Fig. 7 covers another form of the seam somewhat similar to that shown in Fig. 4, in which the hose material 28 is divided with two upstanding ends 21, 21 which may be used stitched at 28, 29 over which is mounted a cover strip 30. The strip 80 may be secured by an additional stitching at 28 if desired or secured in any other suitable manner. In this type the wire 30a may be used

or omitted as found more desirable." Column 2, Lines 35-50) embedded within or layered on the seam ("seam" as seen in Figures 2-4, 6, and 7) to provide rigidity to the gas delivery tube. The resultant effect of the rigidizing element is the stiffening of the seams in order to "prevent transverse collapse of the hose" thereby permitting the orientation whereby "the stiffening wire will tend to hold the hose in the extended or open position" (Column 2, Line 55 thru Column 3, Line 15) and further "facilitates the collapse of the hose in the longitudinal direction" for improved "stowage space" (Column 3, Lines 15-30).

Therefore, it would have been obvious to one having ordinary skill in the art to modify the seam of Ovizinsky to include a rigidizing element in the form of a wire at the seam as taught by Hamblin to impart stiffening of the seams of the air delivery tube during the conveyance of fluid, whilst also permitting the operational collapse of the hose when desired to improve storage capabilities of the air delivery tube.

As to Claim 12, the modified Ovizinsky, specifically Ovizinsky discloses the at least one seam (defined by the stacked configuration as shown in Figures 13 and 14, "The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit." Para 0188) comprises a first seam (one of left/right as shown in the stacked confirmations of Figures 13 and 14) and a second seam (other of left/right as shown in the stacked confirmations of Figures 13 and 14).

As to Claim 13, the modified Ovizinsky, specifically Ovizinsky discloses a first seam (one of left/right as shown in the stacked confirmations of Figures 13 and 14) and a second seam (other of left/right as shown in the stacked confirmations of Figures 13 and 14).

Yet, **does not expressly disclose** "the first seam provides greater or lesser rigidity to the gas delivery tube than the second seam."

Hamblin teaches the implementation of rigidizing element ("wire" as seen in Figures 2-4, 6, and 7) is optional ("the wire 30a may be used or omitted as found more desirable." Column 2, Lines 35-50 and further shown in Figure 5 shows seams with and without the rigidizing element 31).

In light of the ability to optionally choose whether to use or omit the rigidizing element as taught by Hamblin, the ability to achieve the claimed configuration whereby "the first seam provides greater or lesser rigidity to the gas delivery tube than the second seam" is obvious to try choosing from a finite number of identified, predictable solutions with a reasonable expectation of success, whereby success would be defined by the ability to modulate the movement of the gas delivery tube as desired.

It has been held where the general conditions of a claim are disclosed in the prior art discovering the optimal or workable ranges involves only routine skill in the art. Consequently the decision to impart the rigidizing element on one of the seams but not the other, to achieve the claimed orientation of "the first seam provides greater or lesser rigidity to the gas delivery tube than the second seam" appears to be an obvious consideration to one having ordinary skill in the art at the time the invention was made.

Moreover, Applicant has not asserted the specific claimed configuration of "the first seam provides greater or lesser rigidity to the gas delivery tube than the second seam" imparts a particular advantage, solves a stated problem, or serves a particular purpose different from

that of enabling the desired dynamic movement of the gas delivery tube; thus the use of the specific configuration appears to lack criticality in its design.

Consequently, one of ordinary skill in the art would have expected Applicant's invention to perform equally well with the modified Ovizinsky, as the construction would yield the predictable results of providing the gas delivery tube with the desired dynamic movement.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of "the first seam provides greater or lesser rigidity to the gas delivery tube than the second seam" as claimed, a known result effective variable, as taught by Hamblin in order to impart the desired dynamic movement of the gas delivery tube.

As to Claim 14, the modified Ovizinsky, specifically Ovizinsky discloses the in use configuration of each gas delivery tube (x20) configured to extend along a side of the patient's face.

Yet, **does not expressly disclose** "the first seam is configured to be positioned anteriorly and the second seam is configured to be positioned posteriorly with respect to the patient's face."

The explicit placement of the seams with respect to patient's face is obvious to try choosing from a finite number of identified, predictable solutions with a reasonable expectation of success, whereby success would be defined by the ability of the gas delivery tube to have the patient contacting portion (one of 620/621) and the non-patient contacting portion (other of 620/621) being separated by each seam so that the seam with its rigidizing element does not effectuate the patient's comfort during use.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the orientation of the seams with respect to the patient's face as addressed by the modified Ovizinsky, a known result effective variable, in order to achieve the placement of the seam without the rigidizing element effectuating the patient's comfort during use.

As to Claim 15, the modified Ovizinsky, specifically Hamblin teaches the rigidizing element ("wire" as seen in Figures 2-4, 6, and 7) are constructed of metal materials ("Where stiffening is used as shown in the drawings to prevent transverse collapse of the hose, it should be made of a spring wire or strip metal ..." Column 2, Lines 55 thru Column 3, Line 20).

As to Claim 16, the modified Ovizinsky, specifically Hamblin teaches the rigidizing element ("wire" as seen in Figures 2-4, 6, and 7) imparted on a gas delivery tube.

Yet, **does not expressly disclose** "rigidising elements comprise a yarn having a lower melt temperature than surrounding yarn".

In an alternative embodiment of Ovizinsky (as addressed with Figure 80-1 in Paras 0260-0261), Ovizinsky teaches the configuration of a gas delivery tube ("Referring to FIG. 80-1, a process of forming an air delivery conduit according to another example is shown." Para 0260) having a rigidizing element (2035, "support elements 2035 of a relatively lower melting temperature embedded therein to form a support structure 2040 by application of heat. ... The textile substrate 2030 may comprise a flexible textile including primarily fibres/yarns or a sheet material of a certain melting temperature. The embedded support elements 2035 may include fibres/yarns of a lower melting temperature. The support elements 2035 may include polyester, polypropylene or others. The support elements 2035 may be woven, embroidered, weft-inserted, crocheted, braided, felted, fused or knitted into the textile substrate 2030 during or

after the manufacturing process of the flat textile substrate fabric. The flexible textile substrate is preferably air-resistant or air-tight and may include a composite laminate or other textile structure. The textile substrate 2030 (with the embedded support elements 2035) is then inserted into a heated tool or flat-plate press. The thermal processing causes the support elements 2035, which are of a lower melting temperature, to fuse together, and create a level of in-built rigidity within the textile substrate in the form of support structure 2040. The fused support elements 2035 are typically more rigid than the textile substrate 2030 and may be visible on one or both sides of the structure. Also, after the thermal processing, the fused elements may become integral to the overall structure. This embedded textile substrate could then be rolled or folded and then seamed/joined along its long edges to form a compression-resistant flexible tube (in the manner shown in FIGS. 77-1 and 77-2)." Paras 0260-0261) comprised of a yarn ("The embedded support elements 2035 may include fibres/yarns of a lower melting temperature." Para 0260) having a lower melt temperature than surrounding yarn ("The textile substrate 2030 may comprise a flexible textile including primarily fibres/yarns or a sheet material of a certain melting temperature." Para 0260).

The resultant effect of this construction is the ability to provide a rigidized gas delivery tube.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of the rigidizing structure of the modified Ovizinsky incorporate an alternative rigidizing structure in the form of yarn as taught by Ovizinsky to yield a rigidized gas delivery tube.

As to Claim 17, the modified Ovizinsky, specifically Ovizinsky discloses the patient contacting portion (one of 620/621) and the non-patient contacting portion (other of 620/621) each including a first elongate length of material and at least one second elongate material (best seen Figures 8, 9, and 15).

As to Claim 18, the modified Ovizinsky, specifically Ovizinsky discloses the non-patient contacting portion (other of 620/621) having at least one second elongate material (best seen Figures 8, 9, and 15); yet, **does not expressly disclose** "a third elongated length of material".

In an alternative embodiment of Ovizinsky (best seen Figures 53-1 thru 54), Ovizinsky teaches the configuration of a gas delivery tube (x20, specifically 1220, "FIGS. 53-1 to 53-3 and 54 show an air delivery conduit 1220 according to another example of the disclosed technology. In this example, the air delivery conduit 1220 includes a tube 1225 constructed of spacer fabrics and textile cover portions 1231, 1232 (e.g., laminated fabric) that substantially enclose the tube." Para 0224) having an additional layer ("an inner layer 1225(2) (e.g., third layer)" Para 0226) on the non-patient contacting portion which provides for the claimed "third elongate length of material".

Explicitly, Ovizinsky teaches the claimed "third elongate length of material" is constructed of "Spacer fabrics feature two complementary slabs of fabrics with a third layer tucked in between. The third or inner layer can take on a variety of shapes including tubes. The specific arrangement of the third layer can provide for a level of cushioning usually nonexistent in two dimensional fabrics. Advantages of spacer fabrics include: no lamination required if the spacer fabric is coated and/or has a secondary finish; pliable and flexible; can retain original shape; can build in air permeability; stability and/or stretch depending on materials chosen;

and/or insulating." Para 0135; "The tube 1225 is constructed of spacer fabrics including one or more outer layers 1225(1) (e.g., first and second layers) that cooperate to form the wall of the tube and an inner layer 1225(2) (e.g., third layer) supported within the internal passage provided by the one or more outer layers. The tube includes a relatively flat or non-cylindrical cross-section with the inner layer providing an anti-crush or occlusion resistant structure. The inner layer may define one or more passages or lumens through the tube." Para 0226).

The resultant effect of the imparting of the claimed "third elongate length of material" on the non-patient contacting side provides for "inner layer providing an anti-crush or occlusion resistant structure" through "stability and/or stretch depending on materials chosen; and/or insulating".

Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of the non-patient contacting portion of the modified Ovizinsky to include the third elongate length of material as taught by Ovizinsky in order to impart stability and/or insulation to the gas delivery tube.

As to Claim 19, the modified Ovizinsky, specifically Ovizinsky discloses the configuration of the patient contacting portion (one of 620/621) and the non-patient contacting portion (other of 620/621) being constructed of a plurality of layers (650 as laminated to 655, "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." Para 0188).

As to Claim 20, the modified Ovizinsky, specifically Ovizinsky discloses each gas delivery tube (x20) is configured to extend in use from a position on top of the patient's head, along a

side of the patient's face, to a position adjacent the entrance of the patient's airways through the seal forming structure (510 as shown in Figure 6, "As illustrated, a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit." Para 0117).

As to Claim 21, the modified Ovizinsky, specifically Ovizinsky discloses each gas delivery tube (x20) comprises a textile material (655, "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." Para 0188).

As to Claim 22, the modified Ovizinsky, specifically Ovizinsky discloses each gas delivery tube (x20) comprises an inner portion (650, as best seen in Figures 10-15, "film laminate 650" Para 0188) covered by an outer layer (655, as best seen in Figures 10-15, "fabric 655 (FIG. 11)" Para 0188), said outer layer (655) comprising the textile material or foam material of the patient contacting portion (one of 620/621) and/or the textile material or foam material of the non-patient contacting portion (other of 620/621).

As to Claim 23, the modified Ovizinsky, specifically Ovizinsky discloses each gas delivery tube (x20); yet, **does not expressly disclose** "an inner portion covered by an outer layer, said inner portion comprising a foam material and/or spacer fabric".

In an alternative embodiment of Ovizinsky (best seen Figures 53-1 thru 54), Ovizinsky teaches the configuration of a gas delivery tube (x20, specifically 1220, "FIGS. 53-1 to 53-3 and 54 show an air delivery conduit 1220 according to another example of the disclosed technology.

In this example, the air delivery conduit 1220 includes a tube 1225 constructed of spacer fabrics and textile cover portions 1231, 1232 (e.g., laminated fabric) that substantially enclose the tube." Para 0224) having an inner portion ("an inner layer 1225(2) (e.g., third layer)" Para 0226) covered by an outer layer (1250 **OR** 1255, "The textile cover portions 1231, 1232 may be constructed of a laminated fabric (e.g., an inner layer of a film laminate 1250 (e.g., polyurethane or medical grade film) and an outer layer of a textile or fabric 1255 (e.g., synthetic or specified fabric)), each of which may be thermoformed to create its shape and then the textile cover portions may be seam welded to couple the textile cover portions." Para 0225).

Explicitly, Ovizinsky teaches the claimed inner portion ("an inner layer 1225(2) (e.g., third layer)" Para 0226) is constructed of "Spacer fabrics feature two complementary slabs of fabrics with a third layer tucked in between. The third or inner layer can take on a variety of shapes including tubes. The specific arrangement of the third layer can provide for a level of cushioning usually nonexistent in two dimensional fabrics. Advantages of spacer fabrics include: no lamination required if the spacer fabric is coated and/or has a secondary finish; pliable and flexible; can retain original shape; can build in air permeability; stability and/or stretch depending on materials chosen; and/or insulating." Para 0135; "The tube 1225 is constructed of spacer fabrics including one or more outer layers 1225(1) (e.g., first and second layers) that cooperate to form the wall of the tube and an inner layer 1225(2) (e.g., third layer) supported within the internal passage provided by the one or more outer layers. The tube includes a relatively flat or non-cylindrical cross-section with the inner layer providing an anti-crush or occlusion resistant structure. The inner layer may define one or more passages or lumens through the tube." Para 0226).

The resultant effect of the imparting of the claimed "inner portion" provides for "inner layer providing an anti-crush or occlusion resistant structure" through "stability and/or stretch depending on materials chosen; and/or insulating".

Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of the gas delivery tube of the modified Ovizinsky to include the inner portion as taught by Ovizinsky in order to impart stability and/or insulation to the gas delivery tube.

As to Claim 24, the modified Ovizinsky, specifically Ovizinsky teaches, in an alternative embodiment, the configuration of a gas delivery tube (x20, specifically 1220, "FIGS. 53-1 to 53-3 and 54 show an air delivery conduit 1220 according to another example of the disclosed technology. In this example, the air delivery conduit 1220 includes a tube 1225 constructed of spacer fabrics and textile cover portions 1231, 1232 (e.g., laminated fabric) that substantially enclose the tube." Para 0224) having an inner portion ("an inner layer 1225(2) (e.g., third layer)" Para 0226) covered by an outer layer (1250 **OR** 1255, "The textile cover portions 1231, 1232 may be constructed of a laminated fabric (e.g., an inner layer of a film laminate 1250 (e.g., polyurethane or medical grade film) and an outer layer of a textile or fabric 1255 (e.g., synthetic or specified fabric)), each of which may be thermoformed to create its shape and then the textile cover portions may be seam welded to couple the textile cover portions." Para 0225).

With respect to the explicit limitation of Claim 24 to require "the outer layer comprises the textile material of the patient-contacting portion and/or the textile material of the non-patient contacting portion", the applicable outer layer is the textile material of 1255 – "an outer layer of a textile or fabric 1255 (e.g., synthetic or specified fabric)" (Para 0225) in the formation of the gas delivery tube (x20) whereby the patient contacting portion (one of 1231/1232) and

the non patient contacting portion (other of 1231/1232) cooperatively formulate gas delivery tube (x20, "FIGS. 53-1 to 53-3 and 54 show an air delivery conduit 1220 according to another example of the disclosed technology. In this example, the air delivery conduit 1220 includes a tube 1225 constructed of spacer fabrics and textile cover portions 1231, 1232 (e.g., laminated fabric) that substantially enclose the tube." Para 0224).

The resultant effect of the outer layer (1255) is the construction of a cover or encasement that retains the inner portion (1225) – "First and second textile cover portions 1231, 1232 cooperate to form a cover for the tube 1225, i.e., first textile cover portion provides a cover for a portion of the tube circumference and the second textile cover portion provides a cover for the remaining portion of the tube circumference. The textile cover portions 1231, 1232 may be constructed of a laminated fabric (e.g., an inner layer of a film laminate 1250 (e.g., polyurethane or medical grade film) and an outer layer of a textile or fabric 1255 (e.g., synthetic or specified fabric)), each of which may be thermoformed to create its shape and then the textile cover portions may be seam welded to couple the textile cover portions." (Para 0225).

Therefore, it would have been obvious to one having ordinary skill in the art to modify the outer layer of the gas delivery tube of the modified Ovizinsky to include the claimed outer layer of textile material as taught by Ovizinsky to cover or encase the inner portion.

As to Claim 25, the modified Ovizinsky, specifically Ovizinsky teaches, in an alternative embodiment, the configuration of a gas delivery tube (x20, specifically 1220, "FIGS. 53-1 to 53-3 and 54 show an air delivery conduit 1220 according to another example of the disclosed technology. In this example, the air delivery conduit 1220 includes a tube 1225 constructed of spacer fabrics and textile cover portions 1231, 1232 (e.g., laminated fabric) that substantially

enclose the tube." Para 0224) having an outer layer (1255) in the formation the claimed "sleeve", cover or encasement that retains the inner portion (1225) and provides distinction to the patient contacting portion (one of 1231/1232) and the non patient contacting portion (other of 1231/1232) to formulate the gas delivery tube (x20).

Therefore, it would have been obvious to one having ordinary skill in the art to modify the outer layer of the gas delivery tube of the modified Ovizinsky to include the claimed outer layer of textile material to cover or encase the inner portion as taught by Ovizinsky in the formulation of a gas delivery tube.

As to Claim 26, the modified Ovizinsky, specifically Ovizinsky discloses the at least one gas delivery tube (x20) is a pair of gas delivery tubes (x20 left and x20 right as seen in Figure 6).

As to Claim 27, the modified Ovizinsky, specifically Ovizinsky discloses a crown connector (515, "FIG. 6 shows an example of a pair of air delivery conduits communicated with the patient interface. As illustrated, a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit. In this example, each conduit 520 is adapted to extend from adjacent to or under the patient's nose, over the patient's cheeks, between the patient's eye and ear, and terminate at the crown of the patient's head." Para 0117) that is configured to be positioned on top of a patient's head in use to fluidly connect between the pair of gas delivery tubes (x20 left and x20 right as seen in Figure 6).

As to Claim 28, the modified Ovizinsky, specifically Ovizinsky discloses the at least one seam (defined by the stacked configuration as shown in Figures 13 and 14, "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622. During the thermoforming process, the laminate adheres to the fabric giving it air impermeable properties. The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit." Para 0188) is thermoformed or ultrasonically welded ("thermoformed" "RF weld" Para 0188) to join the patient contacting portion (one of 620/621) and the non-patient contacting portion (other of 620/621).

As to Claim 30, the modified Ovizinsky, specifically Ovizinsky discloses positioning and stabilizing structure (Figures 6-15) to provide a force to hold a seal-forming structure (510 as shown in Figure 6, "As illustrated, a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit." Para 0117) in a therapeutically effective position on the patient's head for positive pressure airway treatment of sleep disordered breathing ("The present technology relates to air delivery conduits used in Positive Airway Pressure (PAP) systems for treatment, e.g., of Sleep Disordered Breathing (SDB) with Continuous Positive Airway Pressure (CPAP) or Non-Invasive Positive Pressure Ventilation (NIPPV)." Para 0002) as claimed in Claim 11, and further the

construction of a seal forming structure (510) constructed and arranged to form a seal with a region of the patient's face surrounding an entrance to the patient's airways for sealed delivery of a flow of gas at a therapeutic pressure in cm of water ("In use, the PAP device generates a supply of pressurized air (e.g., 2-30 cm H.sub.2O) that is delivered to the patient interface via the air delivery conduit." Para 0105) above ambient air pressure through the patient's respiratory breathing cycle in use.

Yet, the modified Ovizinsky **does not expressly disclose** the specific therapeutic pressure of "at least 6 cm of water".

In light of the disclosure of the modified Ovizinsky, specifically Ovizinsky disclosing the general operational parameters of PAP devices to include the range of 2-30 cm of water, the specific numerical valuation of the claimed "6 cm of water" would be obvious to try choosing from a finite number of identified, predictable solutions with a reasonable expectation of success, whereby success would be defined by the ability to provide PAP operational pressures to treat the patient's sleep disordered breathing.

In considering the numerical valuation of the claimed "6 cm of water" is within the disclosed operational ranges of 2-30 cm of water as taught by the modified Ovizinsky, it has been held that where the general conditions of a claim are disclosed in the prior art discovering the optimum or workable ranges involves only routine skill in the art.

Moreover, Applicant has not asserted the specifically claimed "6 cm of water" provides a particular advantage, solves a stated problem or serves a particular purpose different from that of being a known numerical valuation within the disclosed operational ranges of 2-30 cm of

water suitable for imparting PAP pressure to a patient suffering from sleep disordered breathing; thus, the use of the specifically claimed "6 cm of water" lacks criticality in its design.

Consequently, one of ordinary skill in the art would have expected Applicant's invention to perform equally well with the modified Ovizinsky, as the specific therapeutic pressure is within the disclosed operational ranges suitable for imparting PAP pressure to a patient suffering from sleep disordered breathing.

Therefore, It would have been obvious to one having ordinary skill in the art to modify the operational range of the PAP device of the modified Ovizinsky to operate at the claimed "6 cm of water", a known result effective variable in order to permit the treatment of a patient suffering with sleep disordered breathing to achieve PAP therapeutic pressures.

5. Claim 29 is rejected under 35 U.S.C. 103 as being unpatentable over Ovizinsky et al. (2014/0102456) in view of Hamblin (2,452,047), as applied to Claim 11, and further in view of Smith (2012/0325219).

As to Claim 29, the modified Ovizinsky, specifically Ovizinsky discloses the at least one gas delivery tube (x20) having a superior tube portion (520(2), "FIG. 6 shows an example of a pair of air delivery conduits communicated with the patient interface. As illustrated, a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit. In this example, each conduit 520 is adapted to extend from adjacent to or under the patient's nose, over the patient's cheeks, between the

patient's eye and ear, and terminate at the crown of the patient's head." Para 0117) configured to extend in use from a position on top of the patient's head to a position alongside of the patient's head or face proximate a crown connector (515 as seen in Figure 6) and an inferior tube portion (520(1), "FIG. 6 shows an example of a pair of air delivery conduits communicated with the patient interface. As illustrated, a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit. In this example, each conduit 520 is adapted to extend from adjacent to or under the patient's nose, over the patient's cheeks, between the patient's eye and ear, and terminate at the crown of the patient's head." Para 0117) configured to extend from a position along a side of the patient's head to a position adjacent the entrance to the patient's airways proximate the seal-forming structure (510 as shown in Figure 6).

Yet, the modified Ovizinsky **does not expressly disclose** "wherein one of the inferior tube portion and superior tube portion has a greater rigidity than the other of the inferior tube portion and superior tube portion."

Smith teaches an alternative positioning and stabilizing structure (Figures 1-4) having at least one gas delivery tube (13A/13B, "Referring to FIGS. 3 and 4, tubing assembly 14 includes left and right side arms 13A, 13B (fluidly coupled to a respective side of coupling component 8), each made up of a number of individual tubing segments wherein the tubing segments include a plurality of flexible bellows segments 18 and a plurality of straight segments 20." Para 0020) having an superior tube portion (20, "a plurality of straight segments 20." Para 0020) located

proximate the crown connector (8, "Delivery conduit 6 is structured to communicate the flow of breathing gas from pressure generating device 4 to patient interface device 10 through coupling connector 8 (the breathing gas enters at the top of the head of patient 1). Delivery conduit 6, coupling connector 8 and patient interface device 10 are often collectively referred to as a patient circuit." Para 0018) and an inferior tube portion (18, "a plurality of flexible bellows segments 18" Para 0020) located proximate the seal forming structure (10, "patient interface device 10 includes a patient sealing element 12, a tubing assembly 14 that is fluidly coupled to both coupling connector 8 and patient sealing element 12." Para 0019) for imparting PAP therapeutic pressure to the patient ("Pressure generating device 4 is structured to generate a flow of positive pressure breathing gas and may include, without limitation, ventilators, constant pressure support devices (such as a continuous positive airway pressure device, or CPAP device), variable pressure devices (e.g., BiPAP.RTM., Bi-Flex.RTM., or C-Flex.TM. devices manufactured and distributed by Philips Respironics of Murrysville, Pa.), and auto-titration pressure support devices." Para 0018).

Regarding the remaining limitations "wherein one of the inferior tube portion and superior tube portion has a greater rigidity than the other of the inferior tube portion and superior tube portion", Smith teaches the superior tube portion – constructed of a rigid straight segments (Para 0020) has a greater rigidity than the interior tube portion – constructed of flexible bellows which "permits the bending and elongation described" (Para 0023).

The resultant effect of the configuration of the varying rigidity of the inferior tube portion as compared to the superior tube portion is to "achieve a proper fit without braking the

airflow seal (within tubing assembly 14 and between patient sealing element 12 and the face of patient 1)." (Para 0023).

Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of the gas delivery tube of the modified Ovizinsky to include a variation in the rigidity of the inferior tube portion to the superior tube portion, as taught by Smith to achieve the proper fitment of the positioning and stabilizing structure without breaking the seal of the seal forming structure.

Conclusion

6. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

a. Barlow et al. (11,759,592) shares a common assignee/inventor with the instant application; however, **at this time** there does not appear to be a double patenting rejection applicable. The closest patent claim to the instant independent claim, Claim 11, appears to be patent claim 15, which depends from claims 11 and 1, and recites the features of the positioning and stabilizing structure having at least one gas delivery tube, wherein the gas delivery tube is configured at the entrance of the patient's airways via a seal forming structure, is formed from a textile material and/or foam material, and includes at least one seam (patent claim 1), wherein the seam includes a rigidizing element embedded within or layered on the seam (patent claim 15). Yet, the parentage of patent claim 15 does not address the features of the patient contacting portion and the non patient contacting portion as required by instant claim 11, nor the

features of the an air impermeable material provided to the textile material as required by instant claim 11. Although patent claim 9 recites the features of the patient contacting and non-patient contacting portions, it is noted claim 9 is not within the familial line of patent claim 15. Thus, there is no double patenting rejection **at this time**.

b. Lewis (4,478,661) discloses a fabric hose constructed with yarn for the conveyance of fluid having seams; yet, **does not expressly disclose** the embedding of a rigidizing element within or along the seams nor the use of an air impermeable material.

c. Frater (2008/0173305) discloses a hose having two ends which are joined at a seam in the formation of the hose to convey fluid; yet, **does not expressly disclose** the embedding of a rigidizing element within or along the seams nor the use of an air impermeable material.

7. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ANNETTE F DIXON whose telephone number is (571)272-3392. The examiner can normally be reached M-F 9-5 EST with flexible hours.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Kendra D Carter can be reached at 571-272-9034. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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